DRIVER DROWSINESS DETECTION ON THE EDGE

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ABSTRACT

Driver drowsiness detection is a key component of modern Advanced Driver Assistance Systems (ADAS), aiming to enhance road safety through timely identification of reduced driver alertness. In this work we present a comparative evaluation of several state-of-the-art deep learning models for drowsiness detection, including convolutional and transformer-based architectures, across multiple public datasets and a custom-compiled dataset. Our analysis focuses on cross-dataset generalization, lightweight preprocessing strategies, and suitability for on-device deployment. Experimental results reveal significant performance variability across datasets, indicating limited generalization beyond the original training domains. We further show that targeted preprocessing, including face detection and Contrast Limited Adaptive Histogram Equalization (CLAHE), consistently improves model performance without increasing training complexity. Although LiteRT conversion was explored to enable mobile deployment, inconsistent performance and incomplete model support stemming from the lack of a unified and reliable conversion framework limited its effectiveness. These findings highlight key challenges in deploying robust, privacy-preserving, and real-time drowsiness detection systems on resource-constrained devices.

Keywords - Automotive, deep learning, driver drowsiness detection, edge AI

1. INTRODUCTION

Driver sleepiness and fatigue are critical factors posing a serious threat to traffic safety and contributing to road accidents. Drowsy drivers often exhibit impaired judgment, poor coordination, slower reaction times, and reduced alertness, all of which substantially increase the likelihood of collisions and their associated consequences, including injuries, fatalities and property damage. According to the US National Highway Traffic Safety Administration reports [1], drowsy driving causes approximately 91000 collisions and 50000 injuries each year in the US alone. In 2023, 633 deaths were reported from drowsy-driving-related crashes. According to European Union EU) analyses, it is estimated that 10-20% of all road accidents are caused due to driver fatigue [2].

In an effort to reduce traffic accidents, the EU adopted the new General Safety Regulation GSR), that establishes mandatory safety systems aimed at supporting 'Vision Zero' [3], that is the goal of near-zero road fatalities and serious injuries by 2050 and the reduction of deaths and serious injuries by 50% by 2030. This regulation covers general and specific vehicle requirements, including Driver Drowsiness and Attention Monitoring (DDAMS), which is a mandatory requirement to all new motor vehicles sold in the EU as of July 2024 [4, 5]. DDAMS is consisted of two layers: drowsiness detection and driver attention monitoring to ensure the driver remains engaged. In response, software providers in corporation with automotive companies are accelerating the development and integration of AI-driven models in their Driver Monitoring System DMS) to enable early detection of drowsiness and timely driver alerts.

In our earlier work on driver drowsiness detection [6, 7] we introduced lightweight rule-based algorithms. In this paper we extend that line of research by incorporating deep learning models, which offer greater robustness and improved adaptability across diverse driving conditions and user scenarios. The remaining of the paper is presented as follows. Section 2 presents the literature review on the latest advancements in drowsiness detection. In Section 3 we describe the methodology that we followed, and in Section 4 we provide some experimental results and compare different models in terms of performance accuracy and inference latency. The challenges that we faced are described in Section 5. Final conclusions and remarks are drawn in Section 6.

2. LITERATURE REVIEW

In our previous work [6, 7] on driver monitoring we employed algorithms based on physiological indicators such as prolonged eye closure (PERCLOS [8]), yawning frequency and head pose. However models relying on these signs often face many challenges due to their rule-based nature. Such approaches may introduce ambiguity or inaccuracy and typically fail to generalize across individuals, as they do not account for inter-subject variability. Consequently a one-size-fits-all approach is generally infeasible, motivating the exploration of more advanced and adaptive methods for reliably assessing driver state and detecting drowsiness.

In this work, we experiment with more recent AI-driven

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methods, including Convolutional Neural Networks CNNs) and Visual Transformers (ViTs). These methods demonstrate high performance capabilities in complex tasks that require adaptability and advanced pattern recognition. Since our application will be deployed on smartphones, that imposes both storage constraints and computation resources limitations. For this reason, our research was focused on lightweight mobile-friendly models.

In [9] authors perform facial landmark feature extraction using the Haar Cascades method [10], before feeding the features to a lightweight CNN for drowsiness detection. Their method was tested on 20 videos from the NTHU dataset [11]. The proposed model, D2CNN-FLD, with a size of 75 KB, scored a validation accuracy equal to 83%. However authors report performance deterioration in cases of face obstruction by sunglasses and bad lighting conditions. Reddy B. et al. [12] propose a two-step process consisting of a joint face detection and alignment method using MTCNN [13], and a drowsiness detection model, which is a 2-stream AlexNet-like model with inputs of left-eye and mouth regions. The authors create a lightweight model of size 10 MB, by 'distilling' [14] a large baseline network into a smaller student model, preserving performance while significantly reducing execution time. The proposed model achieved an accuracy of 89.5% on 3-class classification (alert, yawning and drowsy), on a custom dataset. Phan et al. [15] present an approach incorporating deep neural networks built upon VGG16, InceptionV3, and temporal variants such as LSTM and DenseNet. Their models demonstrate high average accuracy 96%) across diverse scenarios, including conditions with masks and glasses. The model was tested on a custom dataset which consisted of drowsy driver video frames.

The emergence of visual transformers [16] has reshaped the computer vision landscape, enabling new performance capabilities, since they can better learn global representations. Ghanta et al. [17] employ YOLOv5 for face extraction alongside a ViT-based classifier for drowsiness detection, achieving 95.5% accuracy on a custom dataset. However the substantial computational requirements and large parameter counts of ViT architectures make them impractical for mobile deployment. These limitations have driven increasing interest in lightweight and mobile-optimized ViT variants. In [18] authors present MobileViT a mobile-friendly vision transformer that can be applied on different applications for mobile devices. The authors prove that MobileViT significantly outperforms CNN and ViT-based networks across different tasks and datasets, achieving top-1 accuracy of 78.4 % on ImageNet-1k dataset.

3. METHODOLOGY

In this section, we provide a detailed description of the evaluation pipeline that was designed and implemented to assess the performance of multiple pre-trained drowsiness detection models. These models were systematically evaluated on a heterogeneous collection of datasets, each characterized by distinct properties and recording conditions, with the objective of obtaining a robust and comprehensive understanding of model behavior across a wide range of scenarios.

3.1 Data collection

For model training and validation a large amount of related data is required. To this end, we combined data from different sources in order to create a representative dataset. Specifically the dataset is a combination of publicly available datasets, including UTA-RLDD, [19], YawDD [20], AFW [21], IBUG [22], LFPW [23], HELEN [24], NITYMED [25] and more [11, 26, 27, 28, 29, 30, 31]. To prevent potential model bias, we made sure that the dataset is diverse, and equally represented all racial and gender groups. The dataset consists of 16788 images, categorized in three classes: alert, drowsy and yawning (Table 1). We will refer to this dataset as the 'Custom Dataset'.

Table 1 – Custom dataset split in categories

Label	Train	Validation	Test
0 (alert)	5.465	329	330
1 (drowsy)	6.386	277	508
2 yawning)	2.956	161	376
Total	**14.807**	**767**	**1.214**

In addition, we conducted experiments with supplementary data sources. First, we employed the 'Driver Drowsiness Dataset (DDD)' [30], which comprises approximately 41000 images, including 22 drowsy and 19 non-drowsy drivers, depicting drivers in drowsy and alert states. Furthermore, we utilized another dataset [32] containing images of drivers either yawning or not yawning, consisting of 1448 images in total, with 723 labeled as yawning and 725 as not yawning. We will refer to this dataset as 'Yawn Detection Dataset'. Collectively these data sources yielded a corpus of approximately 57000 images, which was used to evaluate the performance of the proposed models.

3.2 Data preprocessing

Our goal was to keep data preprocessing simple, so as to minimize computational overhead and reduce inference latency thereby enabling real-time inference. We experimented with different image preprocessing techniques to find out which one mostly benefits the drowsiness detection model.

Given our objective of evaluating and comparing multiple pre-trained models, we opted to preserve and reuse the exact preprocessing pipelines originally defined by the respective model authors. This original preprocessing is applied consistently both during the training phase and subsequent inference with the current model. By strictly replicating the preprocessing procedures employed in the original studies, we ensure that each model operates under the conditions

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for which it was originally designed, thereby increasing the likelihood that it attains its maximal predictive performance on previously unseen data.

3.3 Models

We tested the performance of eight different AI models and compared it against their reported results. All the models were chosen because in previous work they were trying to address the drowsiness detection task, or a similar one, using different training strategies and characteristics.

The first model evaluated in this study was the CKLDriver model [33]. This architecture was fine-tuned using MobileNetV2 [34] as the backbone for image-based classification, with the objective of inferring a binary state of driver drowsiness (drowsy vs. not drowsy) from a single input image. The model accepts 64×64 RGB images as input, and the use of MobileNetV2 as the base network makes it a strong candidate for assessing whether a low-latency edge-deployable model can achieve adequate performance in drowsiness detection.

The second model evaluated in this study is DozeGuardRLDD (Real-Time Distracted Driver Detection) [35], a vision–language encoder model. This model was fine-tuned from the SigLip [36] architecture, for binary image classification. Specifically it is trained to discriminate between drowsy and non-drowsy individuals depicted in input images, using the SigLip for Image Classification framework. Owing to its real-time inference capability and task alignment, DozeGuardRLDD was evaluated across multiple datasets.

Furthermore, we evaluated a set of three models, referred as Drowsiness Detection, that were fine-tuned on a single dataset specifically for drowsiness detection in driving scenarios. This set comprises three successive versions of the same architecture, with the first model representing the initial fine-tuned variant for this task and the third model corresponding to the best performing iteration. All three models share the same backbone, namely the google/vit-base-patch16-224 architecture [37].

In addition, we evaluated MobileViT [18] as part of our model collection. MobileViT is a general-purpose classification architecture, which in the past was fine-tuned for driver drowsiness detection. It represents a state-of-the-art, lightweight hybrid model that integrates convolutional neural networks with vision transformers [16], thereby achieving a favorable balance between computational efficiency and predictive accuracy. The model exhibits strong generalization performance on an unseen test set, and its design emphasis on edge deployment makes it a strong candidate for integration into our proposed pipeline.

Additionally we evaluated a fine-tuned version of the YOLO classifier (Drowsiness-Detection-Yolo-Cls) [38], originally developed to detect driver drowsiness from images. This model was trained using the Ultralytics framework and exhibits high performance on an unseen test set, making it a reliable tool for safety-critical applications.

Finally another promising architecture that we tested was the mini-Xception [39], a lightweight successor to the original Xception model [40]. In earlier works, Mini-Xception demonstrated high performance accuracy on similar computer vision tasks (like facial emotion recognition [39], [41]), making it a compelling candidate for our application. The success of mini-Xception is mainly due to the use of residual modules [42], and depthwise separable convolutions. Residual connections facilitate faster convergence and improve classification performance [40], while depthwise separable convolutions substantially reduce computational cost, compared with standard convolutions. Another characteristic of the mini-Xception is the elimination of fully-connected layers and their replacement by the global average pooling operation, resulting in a model with approximately 58000 parameters and a size less than 1 MB. These characteristics make mini-Xception particularly well-suited for deployment on hardware-limited platforms such as smartphones. This model was trained from scratch, meaning no pre-trained weights were used for finetuning.

3.4 Pipeline

Our processing pipeline consists of three stages. In the first stage, we extract the data and corresponding labels from all datasets. Subsequently we randomly sample N frames from the combined pool of datasets. The sampling procedure is designed, where feasible, to obtain an equal number of samples from each dataset in order to balance their representation.

After the selection process, each frame was individually provided as input to the corresponding model. Each model applied its designated preprocessing pipeline and subsequently performed inference, allowing it to determine whether the frame exhibited indicators of drowsiness.

Model performance was evaluated on previously unseen data using accuracy and F1-Score as evaluation metrics. Accuracy was chosen due to its straightforward interpretation, measuring the proportion of correctly classified frames relative to the total number of samples, and thus providing a clear indication of the model's overall classification performance. The F1-score is the harmonic mean of precision and recall and is a suitable evaluation metric for classification tasks with imbalanced datasets. It is particularly useful in our problem because precision and recall are in trade-off, and it provides a single measure that balances both.

Extending the initial analysis, the two highest-performing models were converted to the LiteRT[1] format. This format was selected to facilitate deployment on mobile devices, as LiteRT is widely adopted for on-device inference and optimized for resource-constrained environments.

[1] https://ai.google.dev/edge/litert

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Conducting this analysis is particularly important because the conversion process typically involves model compression and optimization, which may lead to a reduction in model size and potentially affect predictive performance. To quantify any performance degradation introduced by the conversion, the first three stages of the evaluation pipeline were repeated using the newly generated LiteRT models in place of the original implementations.

4. RESULTS

This section presents the results obtained from the best performing models and discusses the key observations derived from their performance. This section reports only the most informative findings obtained from the best performing models.

Specifically the CKLDriver model and the three variants of the Drowsiness Detection models were excluded from subsequent analyses. Although these models achieved high accuracy on the test partitions of their respective training datasets, we were unable to replicate comparable performance when they were evaluated on genuinely unseen data. Furthermore, when inference was conducted on our Custom dataset, these models did not exhibit satisfactory generalization capability which motivated their exclusion from the results presented here.

Additionally results obtained from LiteRT for all models were not included in Tables 2 and 3, as we were unable to successfully convert all models to this format and, consequently could not obtain a complete and consistent set of results across all evaluated architectures.

Our initial results in Table 2 show that performance without face cropping varies widely across datasets and is strongly affected by dataset-specific training. YoloClassifier and MobileViT reached perfect accuracy 1.00) on DDD mainly because they were trained or fine-tuned on it; their accuracy dropped sharply on external datasets (Yawn Detection and Custom), revealing poor cross-dataset generalization. DozeGuardRLDD shows a similar pattern, performing well on DDD 0.90) but not on the others. In contrast, mini-Xception delivers more balanced, stable accuracy across datasets, especially on the Custom Dataset 0.91), indicating lower model bias. MobileViT's comparatively strong cross-dataset performance likely comes from being a fine-tuned vision transformer which learns richer more transferable image representations.

The introduction of face detection and cropping as an additional preprocessing step led to improved performance consistency across datasets for most models as it can be seen in the results presented in Table 3. In particular the mini-Xception model exhibited measurable performance gains after face cropping, especially on the Yawn Detection Dataset, indicating that focusing on facial regions enhances its ability to exploit relevant visual cues. MobileViT benefited substantially from this preprocessing strategy as well, achieving a marked increase in accuracy on the Yawn Detection Dataset from 0.65 to 0.96, while maintaining stable performance on the remaining datasets. This improvement further supports the hypothesis that transformer-based architectures with strong global representation capabilities can effectively leverage focused facial inputs. Although YoloClassifier experienced a decrease in accuracy on the DDD dataset following face cropping, its performance on the Custom Dataset remained unchanged. Overall, these results demonstrate that targeted face-centric preprocessing not only mitigates background-induced noise but also enhances the robustness and generalization of models, particularly those designed to reduce dataset bias without introducing additional training overhead.

F1-scores closely followed accuracy trends. Models performing well on their training datasets showed high F1-scores under familiar distributions, while several experienced substantial drops on external datasets, reflecting limited robustness. Mini-Xception maintained the most consistent F1-scores across all datasets (0.91 on the Custom Dataset), with face-cropping further improving scores for both MobileViT and mini-Xception on the Yawn Detection Dataset. Note that metrics on the Custom Dataset were unaffected by the face-cropping step, as all images in that set were already pre-cropped to facial regions.

When deploying the selected models using the LiteRT (TensorFlow Lite) runtime, a significant degradation in predictive performance was observed. Both the mini-Xception and YoloClassifier models exhibited near chance-level accuracy (approximately 0.50), indicating that the conversion and optimization process substantially affected their inference behavior. Despite this overall performance collapse, mini-Xception demonstrated a marginal improvement when face detection and cropping were applied prior to inference, suggesting that isolating the facial region partially mitigates the adverse effects introduced by model conversion. These findings suggest that current LiteRT conversion pipelines affected by limitations such as operator incompatibilities, quantization-induced precision loss, and the absence of a unified conversion framework remain inadequate for reliably preserving the performance of lightweight drowsiness detection models, particularly in safety-critical, on-device deployment scenarios.

Due to the nature of the task, the model should perform well in different lighting conditions, including poor light conditions (for example, at night) and shadows. In an endevour to further improve the mini-Xception model's performance, we experimented with different image preprocessing techniques. We found out that the performance was significantly improved after applying Contrast Limited Adaptive Histogram Equalization CLAHE) [43] on the grayscale versions of the input frames. CLAHE enhances local contrast by dividing an image into small regions and adjusting the contrast within each region independently. This prevents areas of the image from becoming too bright or too dark. As a result, CLAHE can reveal details in

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low-contrast regions without excessively amplifying noise in more uniform areas. This image processing method is particularly useful in driver drowsiness detection, where varying ambient lighting can cause parts of the driver's face to appear too bright or too dark, potentially degrading the model's performance. The effectiveness of CLAHE is clearly illustrated in Figure 1.



Figure 1 – CLAHE applied on dataset image. (Image blurred for identity protection). Image taken from NITYMED dataset, simulating night driving [25].

5. CHALLENGES

As evidenced by the reported results, a major challenge encountered during model evaluation, both in their original implementations and in their mobile device-compatible formats, was the lack of consistent performance across different datasets. Furthermore, the models generally exhibited superior performance on the test splits of the datasets on which they were trained, despite these splits being formally designated as unseen data. This behavior indicates limited generalization capability when the models are applied to external datasets.

Furthermore, based on the limited evaluation performed on the best performing model following its conversion to the LiteRT format, a substantial degradation in performance was observed, making the models not perform better than chance. Although some reduction in accuracy is expected as a consequence of model compression and optimization for deployment on resource-constrained hardware, the magnitude of the observed decline exceeded initial expectations. In view of the safety-critical nature of drowsiness detection systems, only a minor performance loss would be acceptable to support reliable operation in real-world conditions.

In addition, several models could not be evaluated in the LiteRT format due to significant challenges encountered during the conversion process. These difficulties stem primarily from the absence of a unified deployment framework and from limitations in current state-of-the-art libraries supporting model conversion and optimization. Recent trends in AI favor increasingly large and complex models, which typically require substantial computational resources and are most often accessed through server-based architectures via Internet connectivity. While effective in many applications, this paradigm introduces two critical limitations: cloud inference cannot meet the low-latency requirements of a real-time drowsiness system, and transmitting biometric data to external servers raises significant GDPR and privacy concerns.

Table 2 – Performance evaluation of different models on publicly available datasets and on Custom Dataset when not applying face crop.

Model	Dataset	Accuracy	F1-score
3*YoloClassifier	DDD	**1.00**	0.76
	Yawn Detection Dataset	0.55	0.58
	Custom Dataset	0.69	0.58
3*mini-Xception	DDD	0.68	0.68
	Yawn Detection Dataset	0.52	0.86
	Custom Dataset	**0.91**	**0.91**
3*MobileViT	DDD	**1.00**	**1.00**
	Yawn Detection Dataset	**0.65**	**0.96**
	Custom Dataset	0.73	0.73
3*DozeGuardRLDD	DDD	0.9	0.9
	Yawn Detection Dataset	0.6	0.6
	Custom Dataset	0.58	0.55
3*YoloClassifier LiteRT	DDD	0.5	0.33
	Yawn Detection Dataset	0.5	0.33
	Custom Dataset	0.5	0.33
3*mini-Xception LiteRT	DDD	0.5	0.33
	Yawn Detection Dataset	0.5	0.33
	Custom Dataset	0.5	0.33

6. CONCLUSION

Our comparative evaluation demonstrates that no single model generalizes reliably across datasets, underscoring the importance of training on diverse, representative data as a prerequisite for real-world drowsiness detection deployment.

Furthermore, by incorporating an additional yet computationally-efficient preprocessing step, specifically face detection and localization, we consistently improved

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Table 3 – Performance evaluation of different models on publicly available datasets and on Custom Dataset when applying face crop.

Model	Dataset	Accuracy	F1-score
3*YoloClassifier	DDD	0.76	0.76
	Yawn Detection Dataset	0.58	0.58
	Custom Dataset	0.69	0.58
3*mini-Xception	DDD	0.69	0.68
	Yawn Detection Dataset	0.86	0.85
	Custom Dataset	**0.91**	**0.91**
3*MobileViT	DDD	**1.00**	**1.00**
	Yawn Detection Dataset	**0.96**	**0.96**
	Custom Dataset	0.73	0.73
3*DozeGuardRLDD	DDD	0.91	0.9
	Yawn Detection Dataset	0.6	0.6
	Custom Dataset	0.58	0.55
3*YoloClassifier LiteRT	DDD	0.5	0.33
	Yawn Detection Dataset	0.5	0.33
	Custom Dataset	0.5	0.33
3*mini-Xception LiteRT	DDD	0.53	0.40
	Yawn Detection Dataset	0.64	0.60
	Custom Dataset	0.62	0.55

model performance across all evaluated datasets. This finding indicates that, in conjunction with other preprocessing operations e.g. grayscale formatting, pixel value normalization), isolating the facial region is a critical step for enhancing classification accuracy without necessitating the collection of additional training data, which would increase both cost and training complexity. By minimizing the presence of irrelevant visual information, particularly background regions, this approach enables the models to focus on features that are most informative for drowsiness detection. In addition, the application of CLAHE further improved image quality under suboptimal lighting conditions, contributing to more robust and reliable model performance.

Finally based on our results we highlighted some critical challenges in deploying such models either as standalone models or as part of an on-device processing module. We emphasized the need for a unified framework to responsibly convert such models while preserving most of their performance, so they can perform on par with their pre-trained versions. Moreover this conversion is needed in order to avoid sending sensitive information outside the device and on device processing is possible.

As future work, we aim to fine-tune a custom drowsiness detection model on a diverse collection of datasets incorporating the preprocessing steps established here. We will also explore video-based spatio-temporal models to enable real-time inference while further reducing dataset-specific bias.

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